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Histology of the Heart

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Session Objectives

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- 1. Students will be able to describe the three layers of the heart and regional variations with regard to morphology and structural significance.
- 2. Students will be able to describe morphology and structural significance of cardiac valves.
- 3. Students will be able to discern regional differences in histology/function of the cardiac conduction system.
- 4. Students will be able to describe morphology and function of components of the cardiac skeleton.
- 5. Students will be able to describe morphologic composition and functional significance of the pericardium.

Source: Chapter 13, pp 400-408, 432-433. Cardiovascular System, Histology: A text and Atlas, Ross and Pawlina,

How does cardiac structure (histology) affect pathophysiology?

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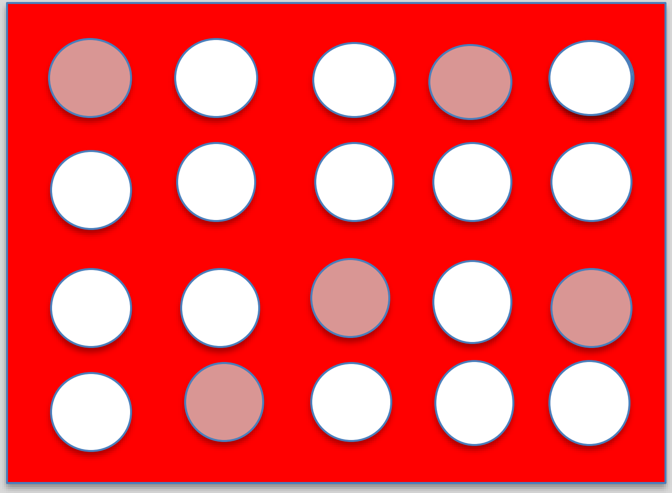
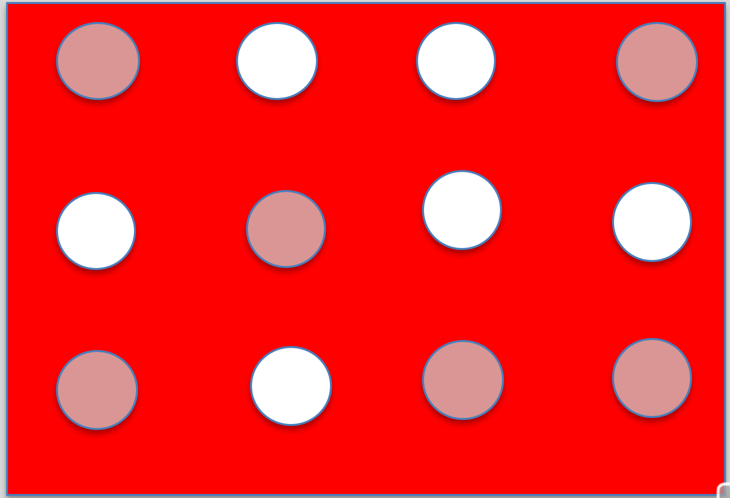
Cross sections of LV myocardium. Red areas packed with myocytes. Circles are capillaries. Myocytes and capillaries run in parallel. Pink = perfused, white = not perfused. LV normally has about 4,000 capillaries/mm^2 (left). Hypertension (right) > myocytes enlarge and push capillaries apart > ~2,500 capillaries/mm^2.



Normal. ~1/4 capillaries perfused at rest. High capillary reserve.



Hypertension. ↓ capillaries, but more perfused (~1/2) to meet ↑ energy needs > ↓ capillary reserve > ↓ exercise capacity with ↑ chest pain on exertion.

Normal heart data

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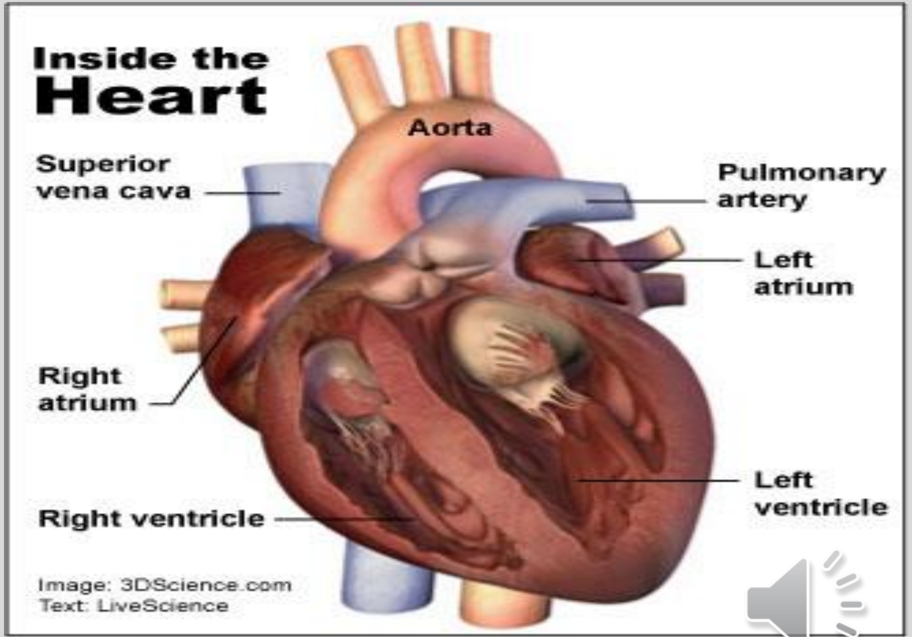
- 250 - 300 gms (females)
- 300 - 350 gms (males, larger hearts due to larger body mass, the major regulator of heart mass)
- Left ventricle wall thickness: 1.3 – 1.5 cm
- Right ventricle wall thickness: 0.3 – 0.5 cm



Source: Chapter on Clinical Anatomy of the Heart from "Classic Teachings in Clinical Cardiology", Ed. MA Chizner, 1996, Laennec Publishing Inc.

Interesting heart and vascular information

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- Mammalian ventricles have ~25 M myocytes/gm (~7.5 B total in human heart)
- Myocyte proliferation ceases shortly after birth
- Continued heart growth is due to myocyte hypertrophy with no "proven" regeneration after this time. Non-myocytes continue to proliferate.
- There is considerable variation in myocyte size, but mean myocyte size in mammals is similar, with more myocytes in species with larger hearts.
- Males and females have about the same number of myocytes, but myocytes are slightly larger in males due to larger body weight.
- Approx. resting heart rates in beats/min (bpm): large whale (10), humans (70), dogs (90), rats (450), mice (600), hummingbird (1200).
- LV systolic pressure in giraffe: ~350mm Hg
- Giraffe's also have valves in their carotid arteries

Source: Mostly studies by AM Gerdes.

Inside the Heart of a Blue Whale

Source: Pinterest

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After boiling, the heart can be blunt dissected into one long muscle.

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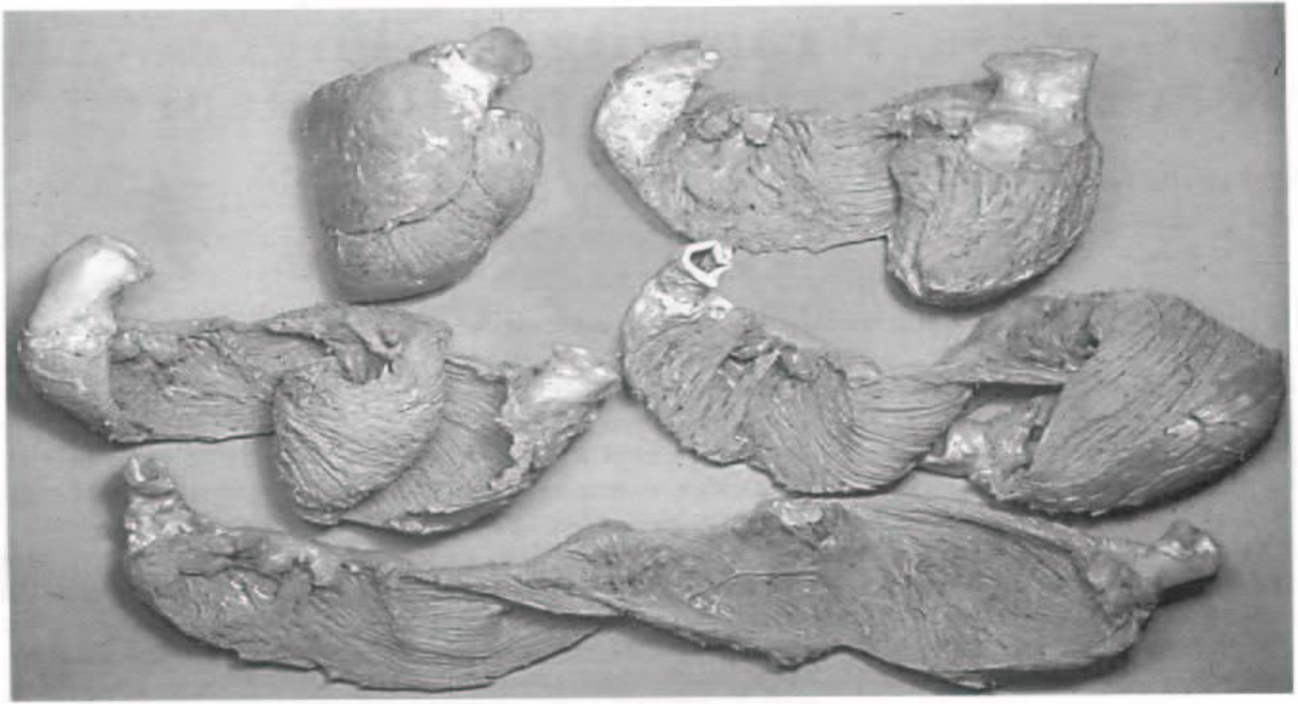

Source: 2008 J Thoracic CV Surgery, G Buckberg.

Heart layers across the wall

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Endocardium

Endothelium, collagen, fibroblasts, smooth muscle, nerves, purkinje cells)

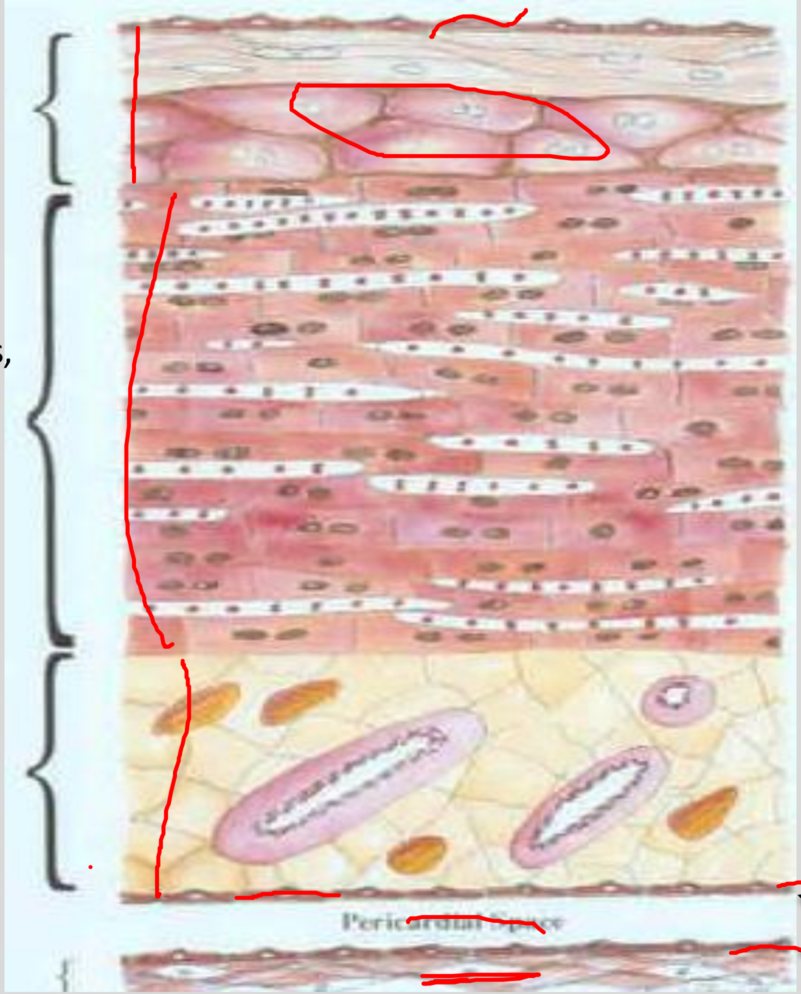
Myocardium

Myocytes, capillaries, blood vessels, small nerves, lymphatics, minimal CT mostly around vessels and between myocytes

Thickness: LV > RV > atria

Epicardium

Fat cells, coronary vessels, nerves, autonomic ganglia



Mesothelium

Pericardium (dense CT)

Source: Gerdes

Pericardium

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- Dense fibrous CT lined by mesothelium on inner surface.
- Acutely, pericardium does not expand (stretch).
- So, increased pericardial fluid can restrict heart function (cardiac tamponade).
- Examples: Pericarditis or chest trauma could fill pericardial cavity with blood).
- Note: First successful open heart surgery performed in Chicago in 1893. Dr Daniel Williams was African American and also established the first black owned hospital in the US. Knife wound to chest from bar fight.
- With chronic cardiac hypertrophy, pericardium can expand slowly, as you likely noted in gross anatomy lab.

Atrial Wall

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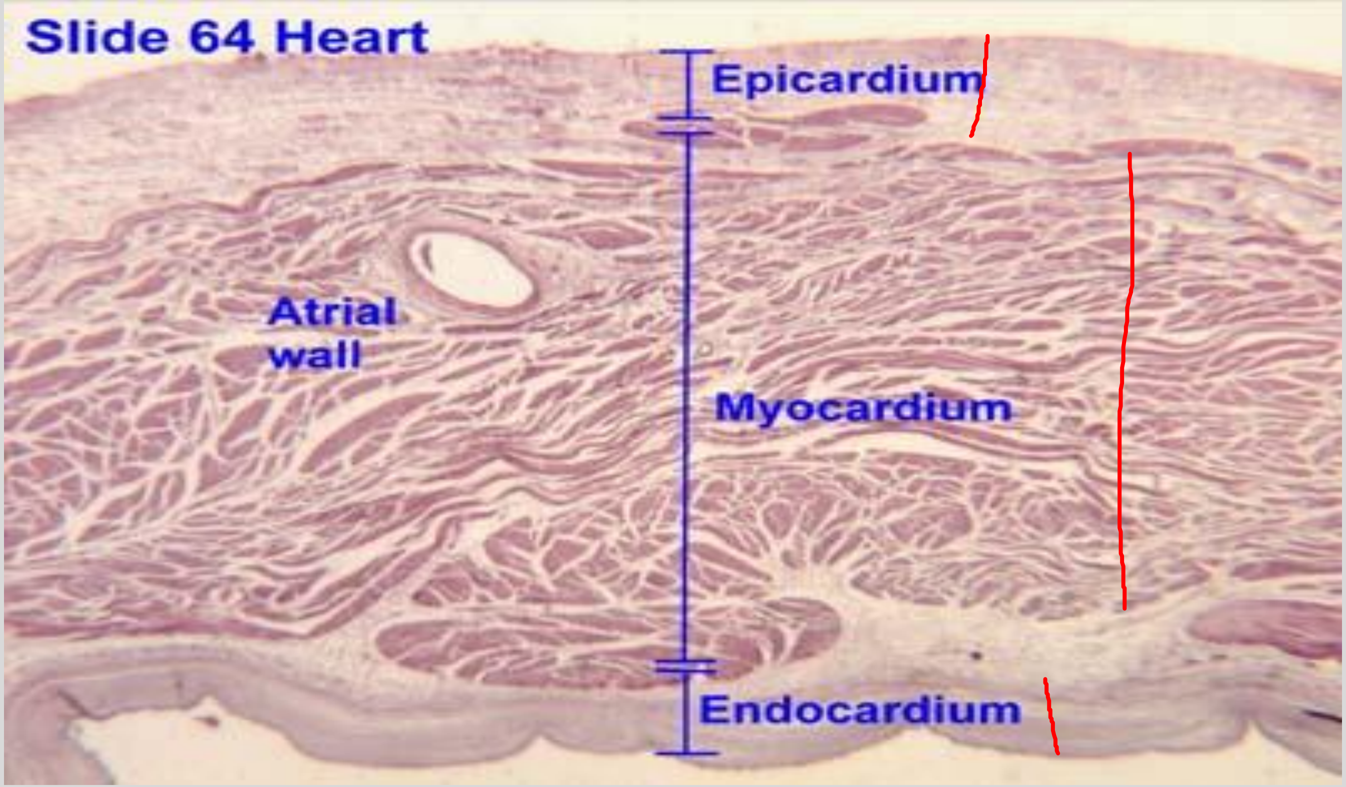
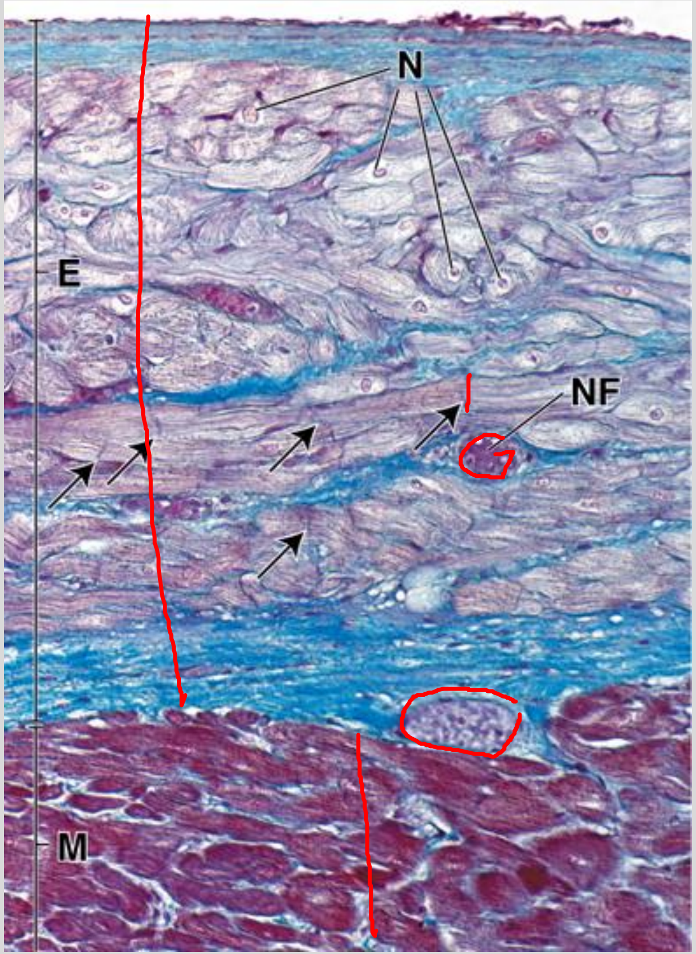


Source: Gerdes

Endocardium



Ventricular cavity at top lined by endothelium. Blue staining is mostly collagen in subendothelial layer and throughout endocardium (E).

Upper 2/3 of image is mostly Purkinje cells. N, nuclei; NF, nerve fibers; arrows, intercalated discs.

Lower 1/3 is myocardium (M)

Source: Ross and Powlena 13.10

Endocardial Surface: Atria and Ventricle

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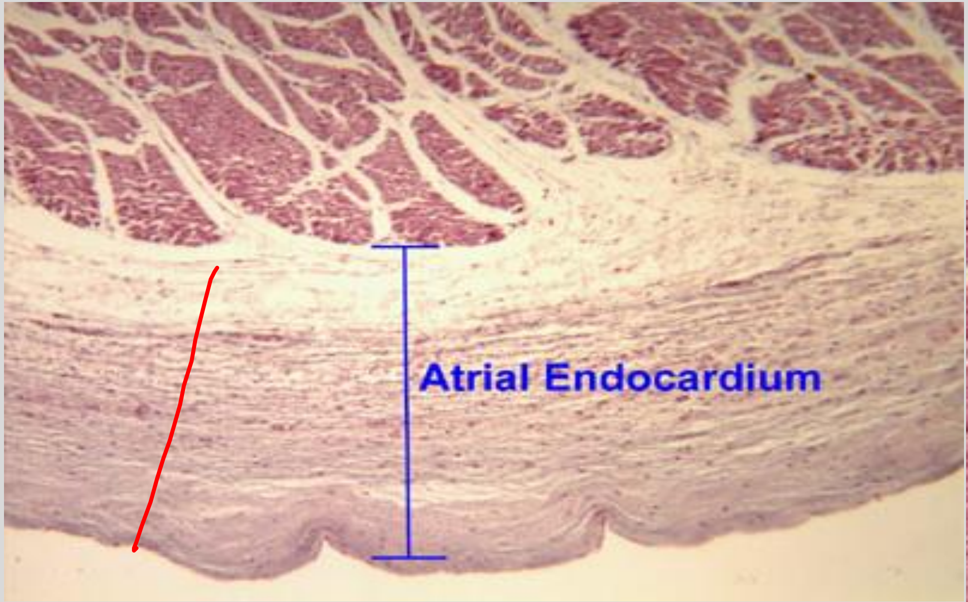
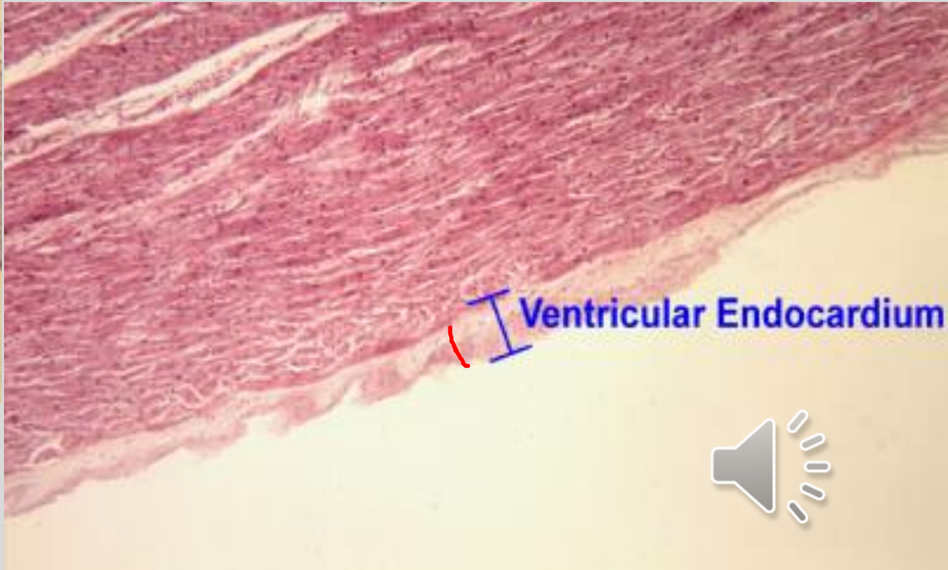
Atria

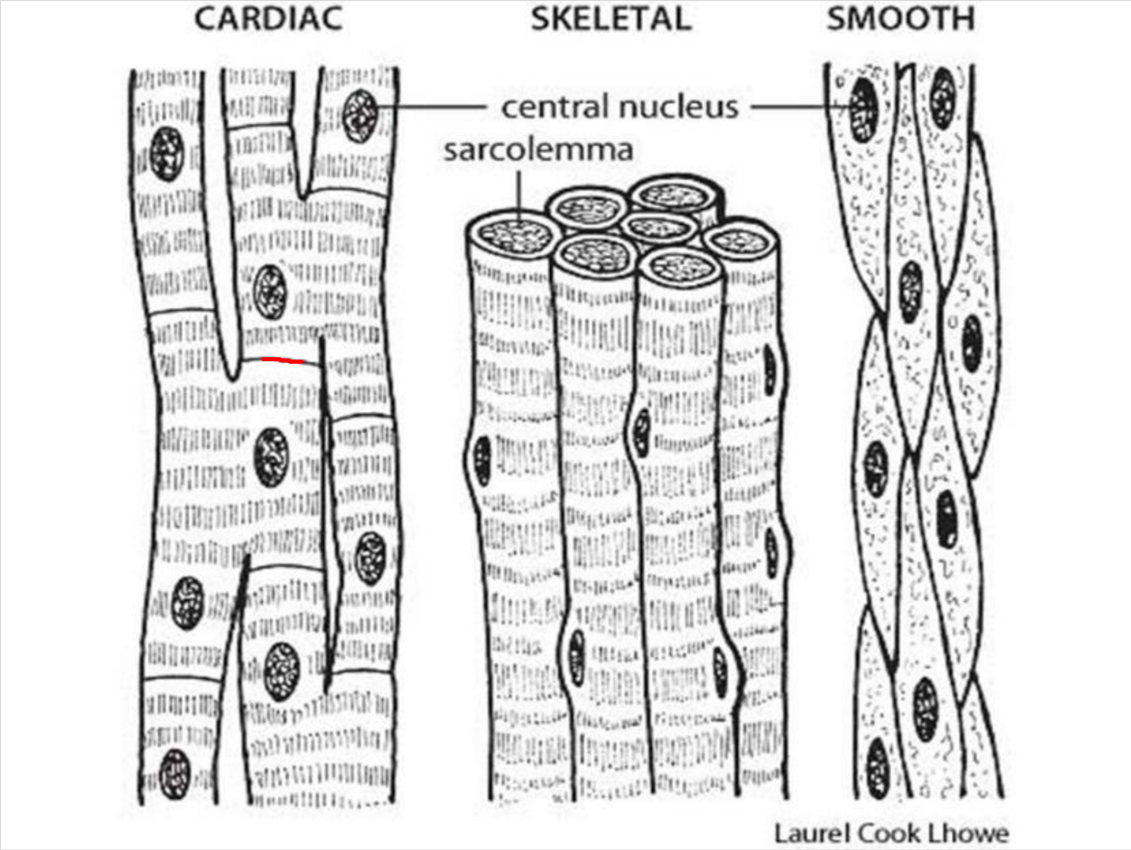


Ventricle



Source: Google images

Muscle types



A Clinical Overview of Muscle Types: dummies.com

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Mammalian isolated cardiac myocytes

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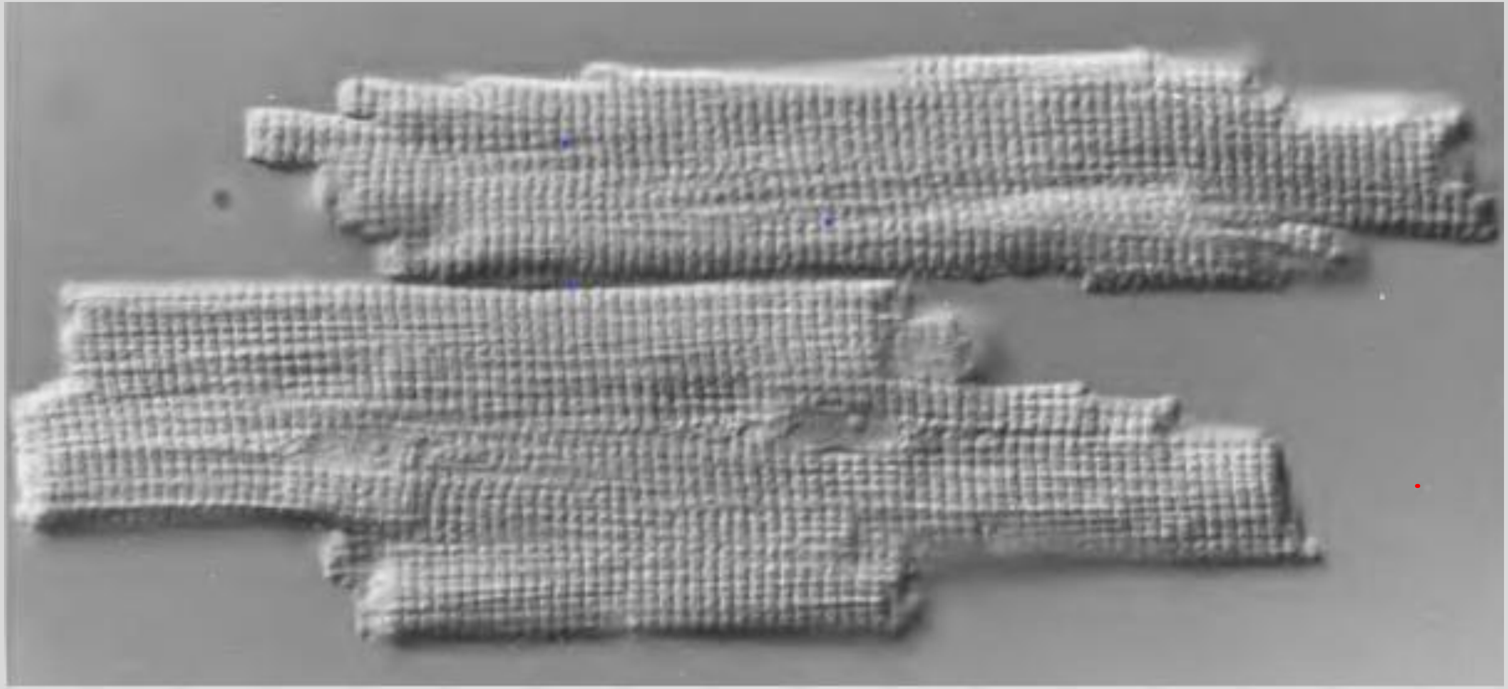
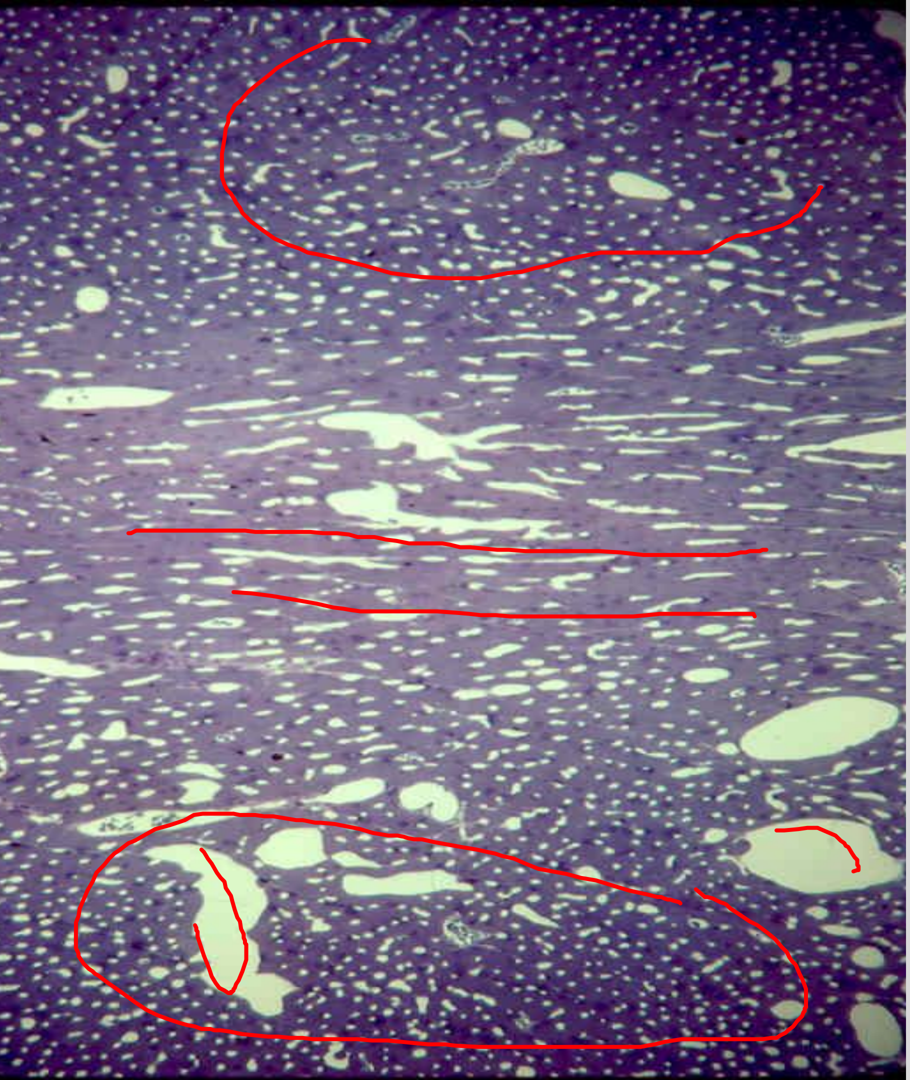


No two myocytes are exactly alike. Shape is complex but ~ that of a cylinder. Branching occurs at the ends of cells (IDs). A given myocyte connects with many others. Y shaped myocytes are rare.



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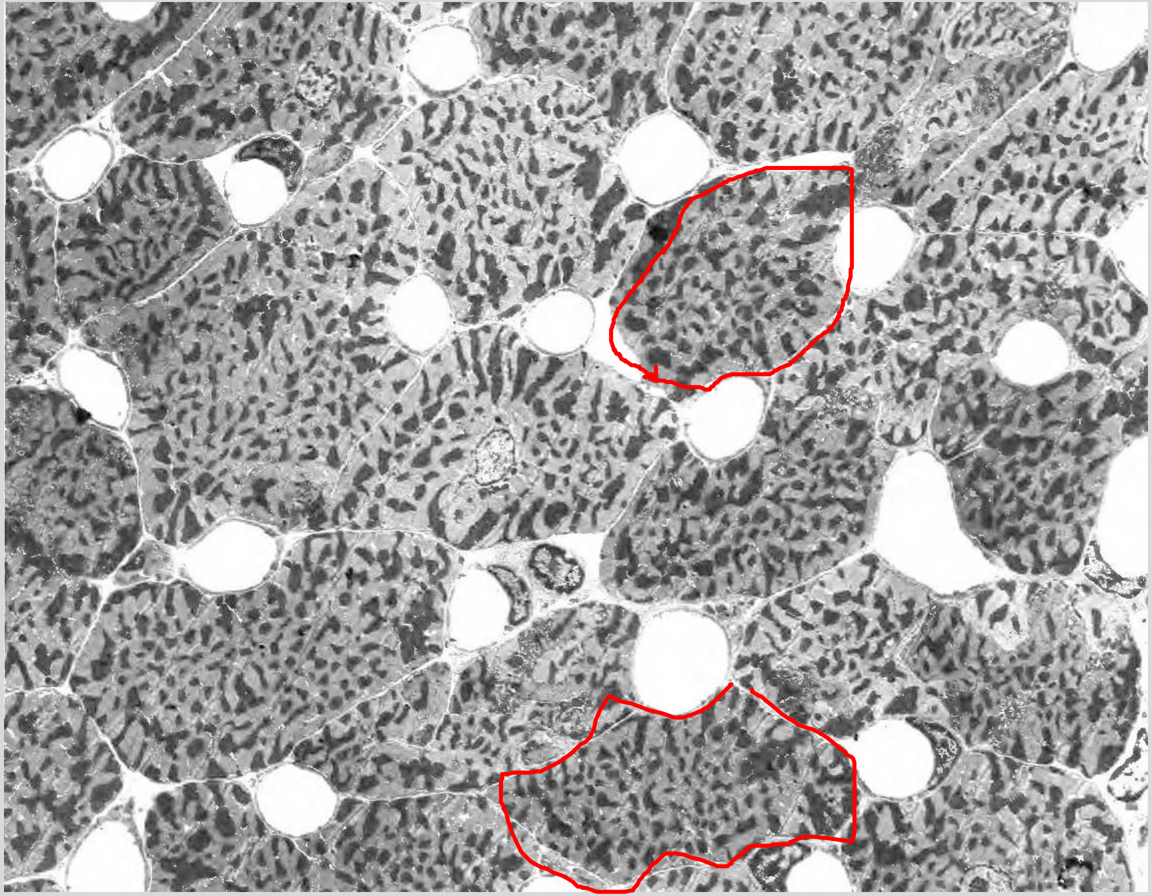
Myocytes change direction across wall somewhat like a Japanese fan (180 degrees from endo- to epimyocardium). Transmural section of rat RV perpendicular to base-apex axis.

Epicardial surface at bottom, **endocardial surface at top**. Myocytes (purple) and capillaries (small round white profiles) at top and bottom are cross-sectioned. Myocytes in middle are longitudinal or obliquely sectioned. Note: Rat heart is thin enough to show this in one field of view. Small white spots are patent capillaries. In adults, the ratio of capillaries/myocytes is approximately 1:1 (also true in humans).

Source: Gerdes

TEM Rat LV Myocardium (Cross Section)



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Capillaries (clear circles) and myocytes (darker cells) cut in cross section. Darkest structures in myocytes, mitochondria, occupy ~1/3 of myocyte volume. Grey areas in myocytes are myofibrils. A central nucleus can be seen in some myocytes.

Source: AM Gerdes

Normal human heart showing branching myocytes (SEM)

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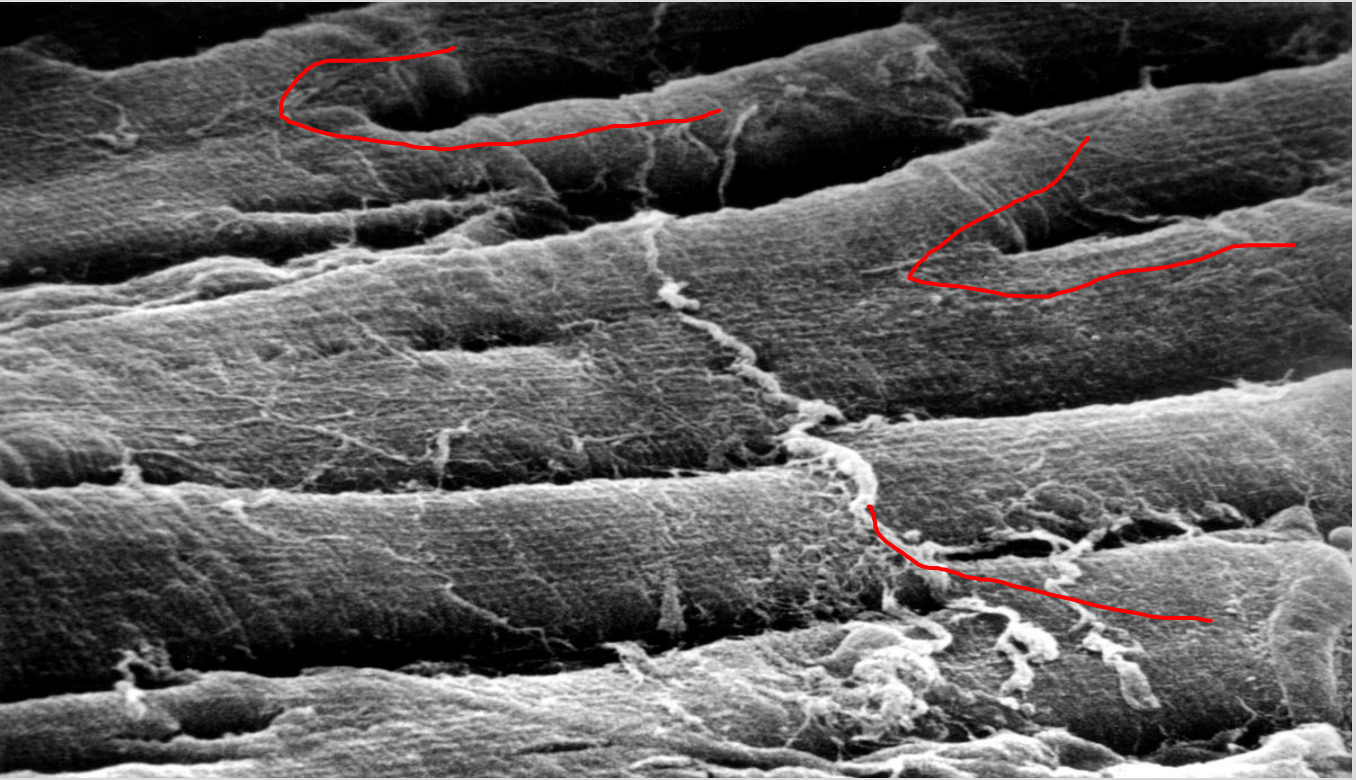

Source: AM Gerdes and heart transplant program at USF, Tampa

Cardiac Myocytes

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Intercalated discs join cells, help synchronize myocyte contraction by allowing electrical coupling with relatively unrestricted ionic passage across the membranes of adjoining cells.

Cardiac muscle is specialized striated muscle, has more mitochondria (25-35% of cell) than skeletal muscle (3-5% of cell), and is very dependent on aerobic metabolism.

Source: Ross

Cardiac Muscle: Routine H&E

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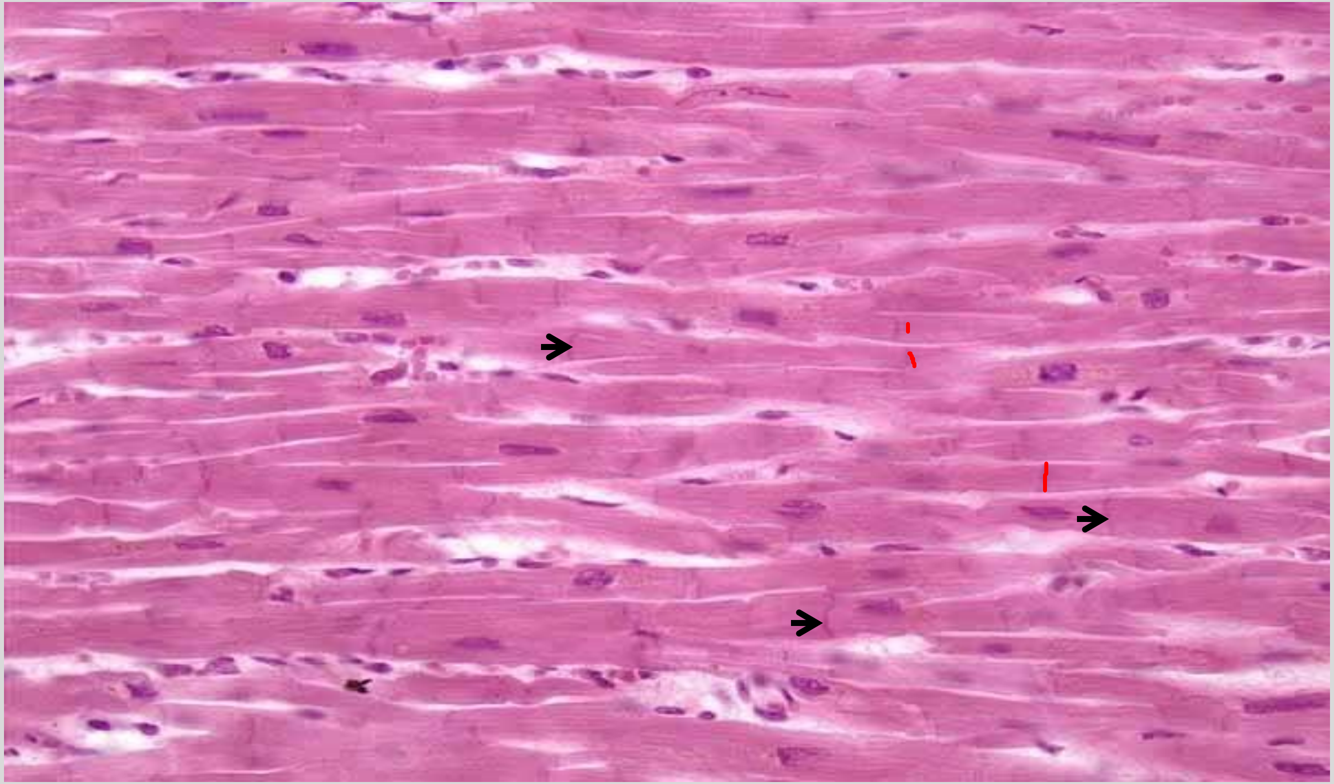
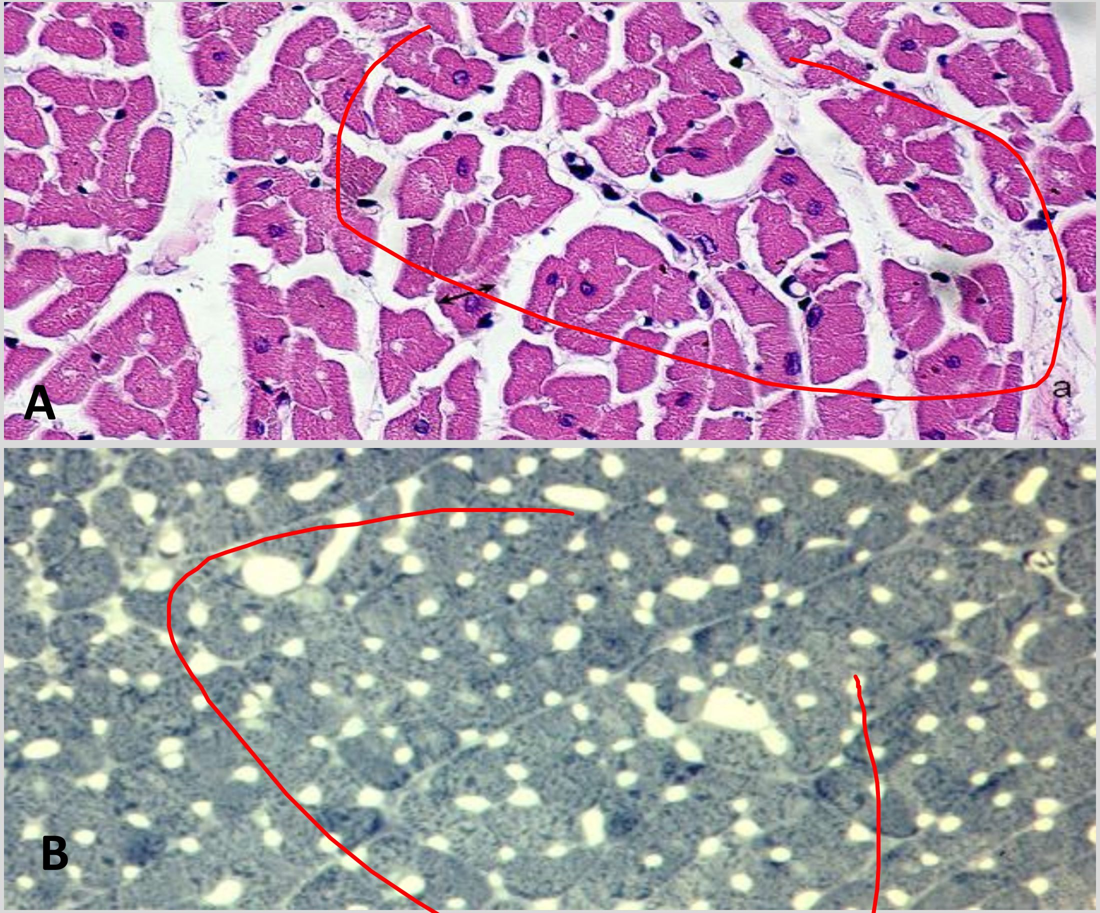


Arrows indicating intercalated discs

Source: Anatomyforme.blogspot.com

Transversely sectioned myocardium



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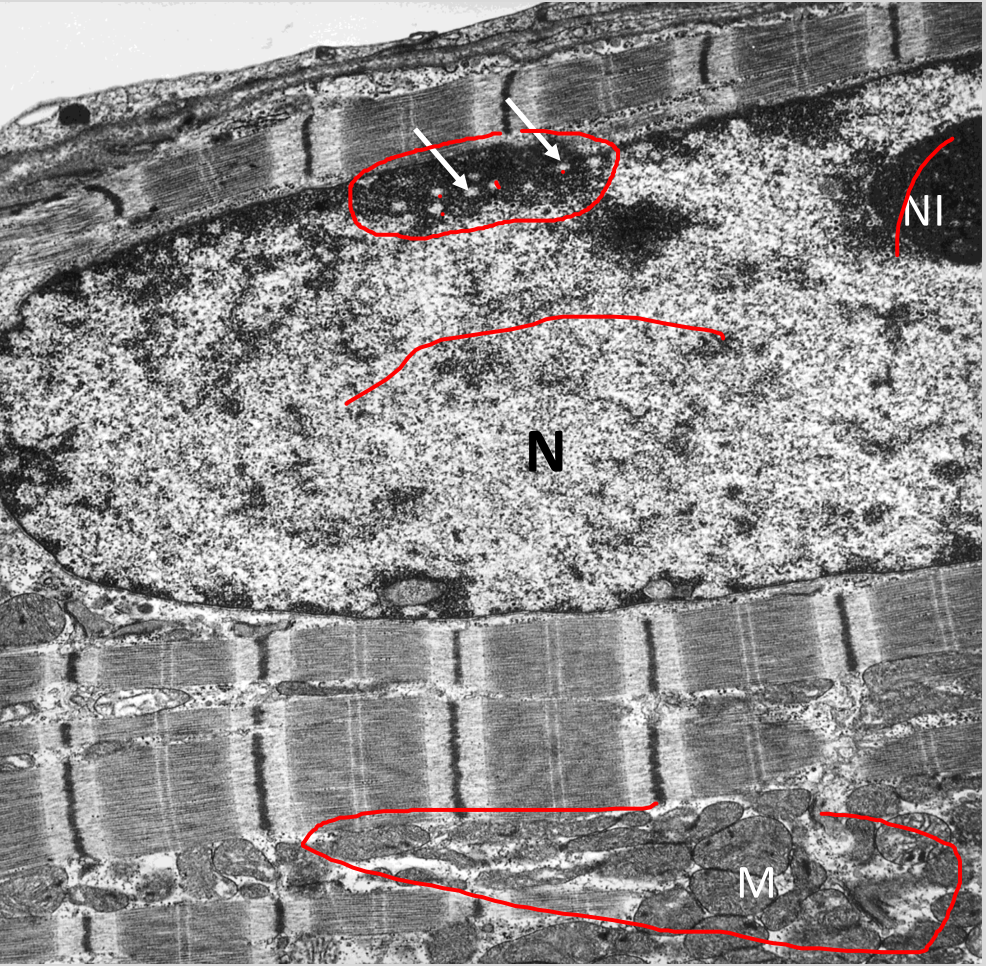
A. Typical H&E prep from immersion fixed tissue.

B. Optimum preservation. Perfusion-fixed tissue with better preservation of true structure. **Myocytes (purple) are tightly packed and borders difficult to discern**. The high degree of vascularization is also better appreciated with improved prep. **Note extensive capillaries** (white circles).

Human Heart



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Likely best TEM example of optimally fixed normal human heart (perfusion fixed with glutaraldehyde). Note nucleus (N), nucleolus (NI), nuclear pores (arrows), mitochondria (M).

Source: Gerdes AM. Cardiomyocyte Ultrastructure. In “Muscle: Fundamental Biology and Mechanisms of Disease”. 2012.

Papillary Muscles

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- -Fingerlike projections from the inner surface of the heart connecting to chordae tendinae
- -Parallel myocytes and capillaries form the core
- -Covered by endocardium on outer surface
- -Have been used by animal physiologists to study cardiac muscle function *in vitro (less complicated since fibers run in same direction)*

LV of rat showing nerves (TEM)

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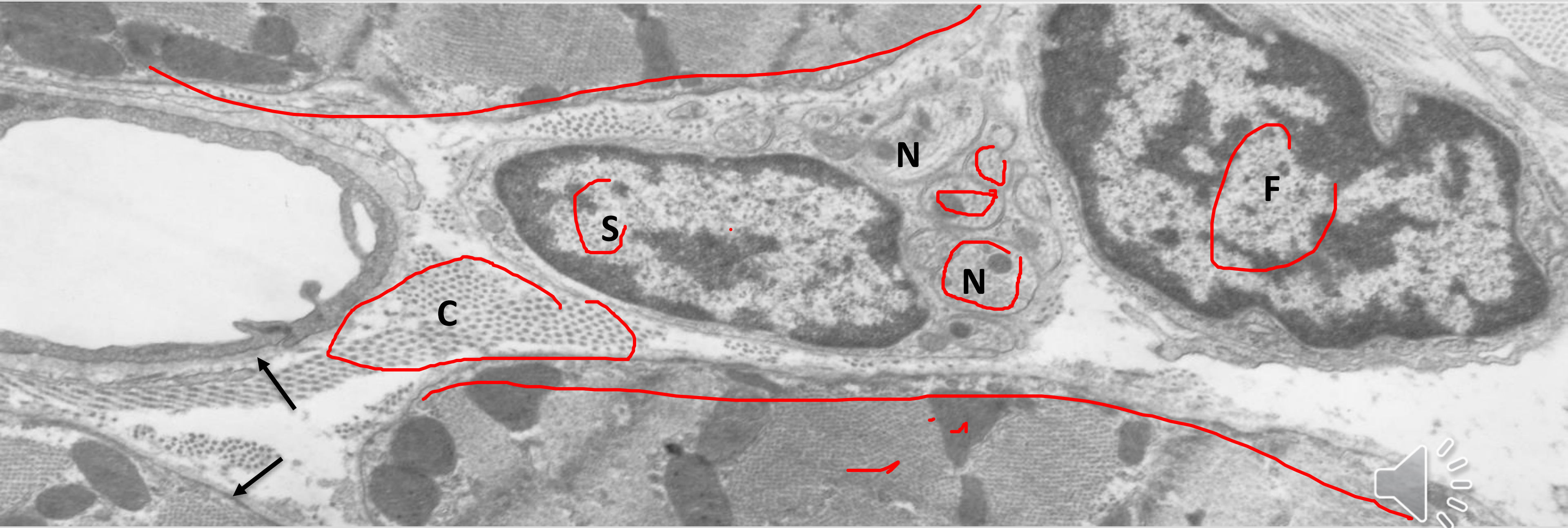
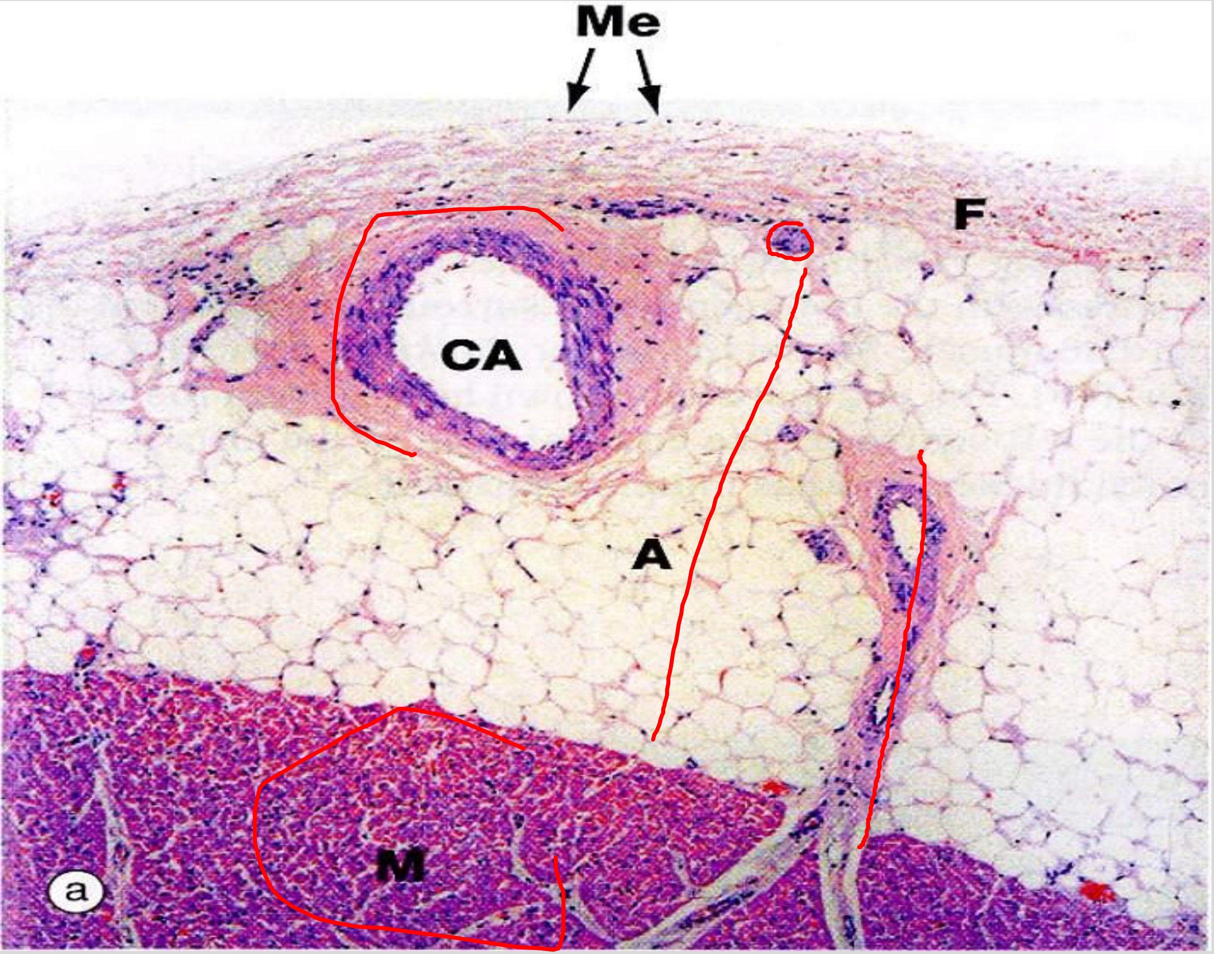


Schwann cell (S), nerve fibers (N) with mitochondria and neurofilaments, collagen fibers (C), fibroblast (F). External lamina (arrows) around myocytes and endothelial cells but not fibroblasts. Source: AM Gerdes

Epicardium



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***Epicardium*:**

-simple squamous epithelium or mesothelium (Me)

-elastin/collagen (F)

- Coronary artery (CA)

-adipose tissue (A)

***Myocardium*:**

-Cardiac myocytes (M)

Source: Quizlet

Epicardium

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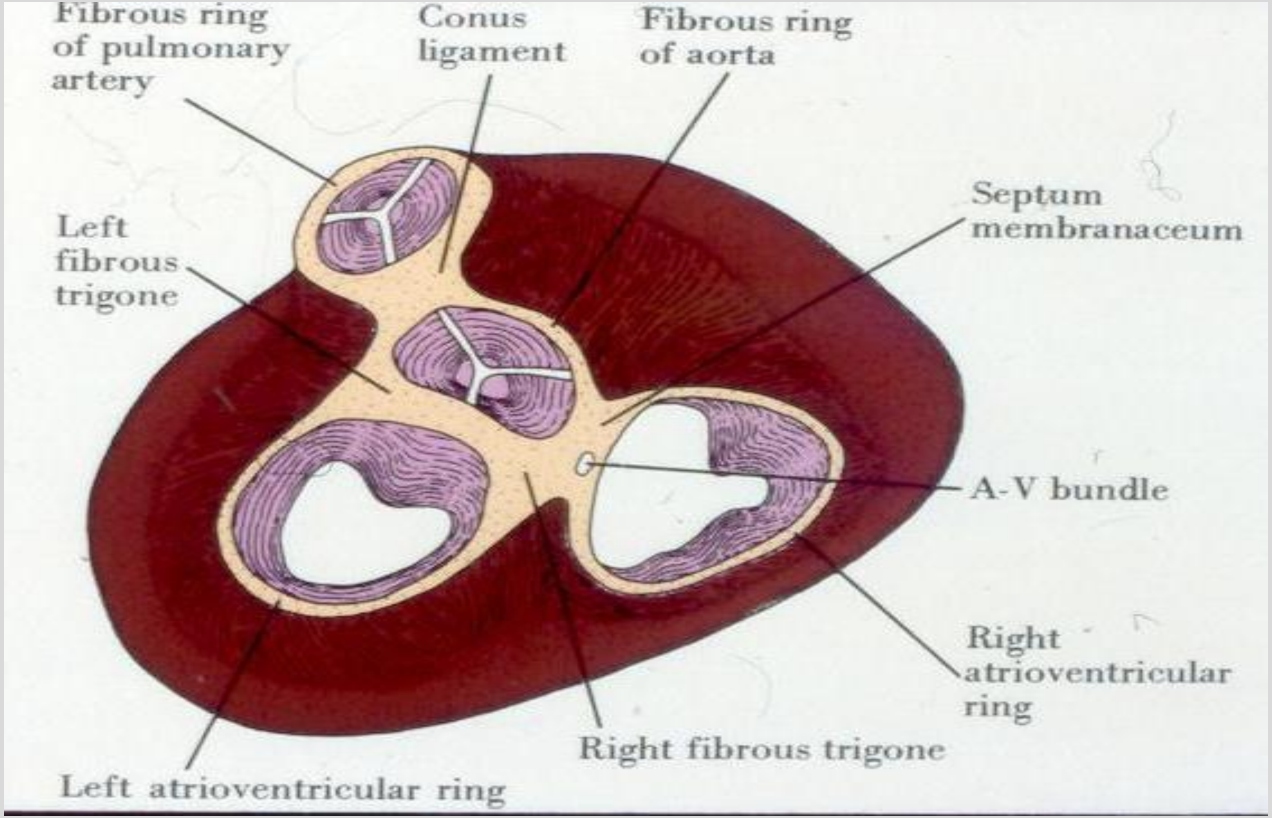
- **Layer of mesothelial cells** on outer surface of heart with underlying connective tissue.
- **Blood vessels and nerves** (autonomic w postganglionic parasympathetic ganglia) which supply the heart are present in the epicardium
- Largely composed of **adipose tissue** which acts as cushion in the pericardial cavity.

Cardiac Skeleton



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Dense fibrous **CT surrounding 4 major valves**, between valves **(trigones)**, and in membranous (upper) **interventricular septum.**

Source: Images.search.yahoo.com

HEART VALVES and chordae (simplified)

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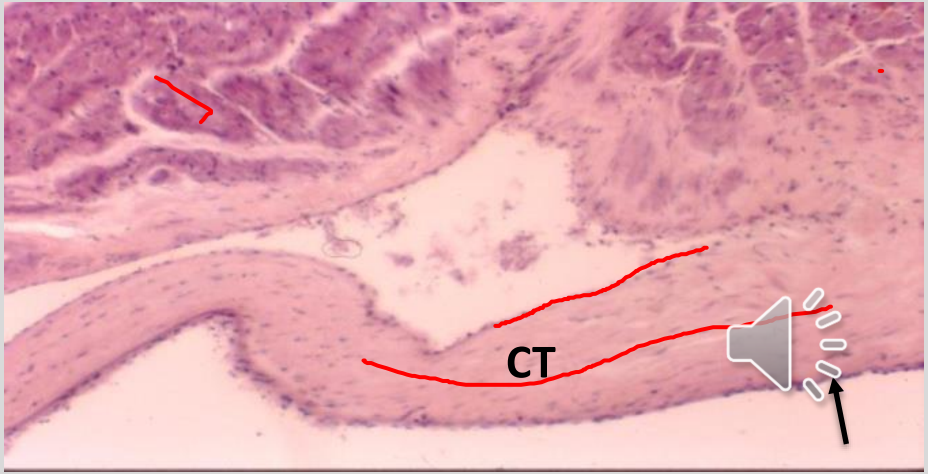
- Valves: lined by simple squamous epithelium (endothelium, arrow) on both sides with dense connective tissue (CT) core extending from fibrous ring .
- Chordae tendinae: dense CT core covered by endothelium. (not shown)

- Avascular:
- get nutrition from blood
- circulating within chambers



Source: Gerdes

Chordae Tendinae

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Dense collagen core covered by endothelium with no direct vascular blood supply.

Source: Univ Michigan virtual histology

Conduction System

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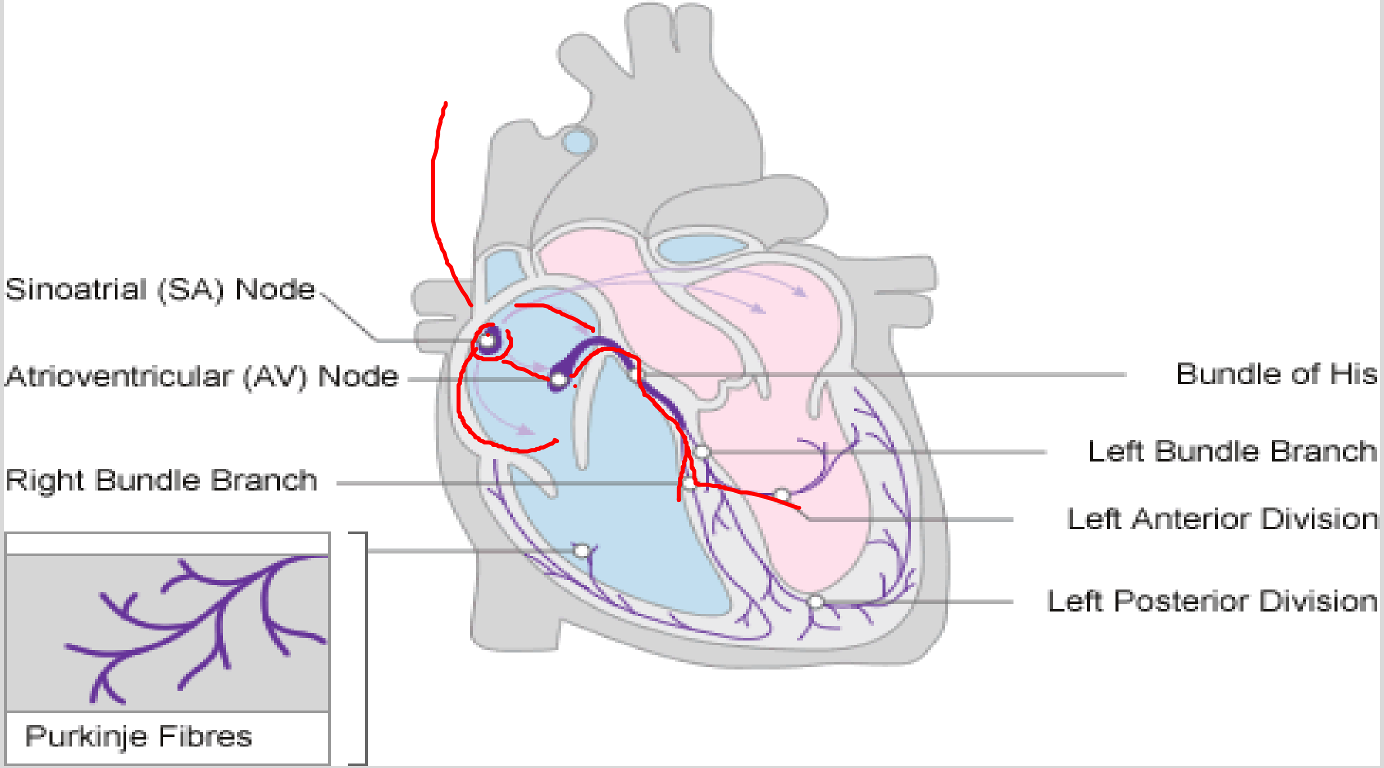

Source: http://www.nottingham.ac.uk/nursing/practice/resources/cardiology/function/conduction.php

<u>**CARDIAC CONDUCTION SYSTEM** (overview)</u>

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- Electrical impulses generated at **sinoatrial (S-A)** node – group of specialized cardiac muscle cells present near junction of superior vena cava and right atrium: "**pacemaker**" of the heart: 60 – 100/min.
- The impulse which the S-A node initiates spreads along the cardiac muscle fibers of the atria and along **internodal tracts** made up of <u>modified</u> cardiac muscle fibers, then picked up at the **A-V node** and conducted across the fibrous skeleton to the ventricles by the **Bundle of His.**
- The bundle of His divides into **right and left bundle branches** and then into subendothelial branches called **Purkinje fibers**

Histology of Conduction Cells (overview)

- **Nodal cells, A-V bundle, bundle branches, Purkinje fibers**: Modified cardiac muscle cells specialized to conduct impulses.
- The **cells in the SA and AV nodes** are **smaller** than normal working myocytes. **Purkinje fibers, AV bundle, and the bundle branches are larger** than normal myocytes. Conduction cells have **fewer myofibrils, few or no T-tubules** and are **rich in glycogen**. The parts of the cardiac conducting system convey the impulse about 4 X faster than regular cardiac muscle fibers.

SA Node

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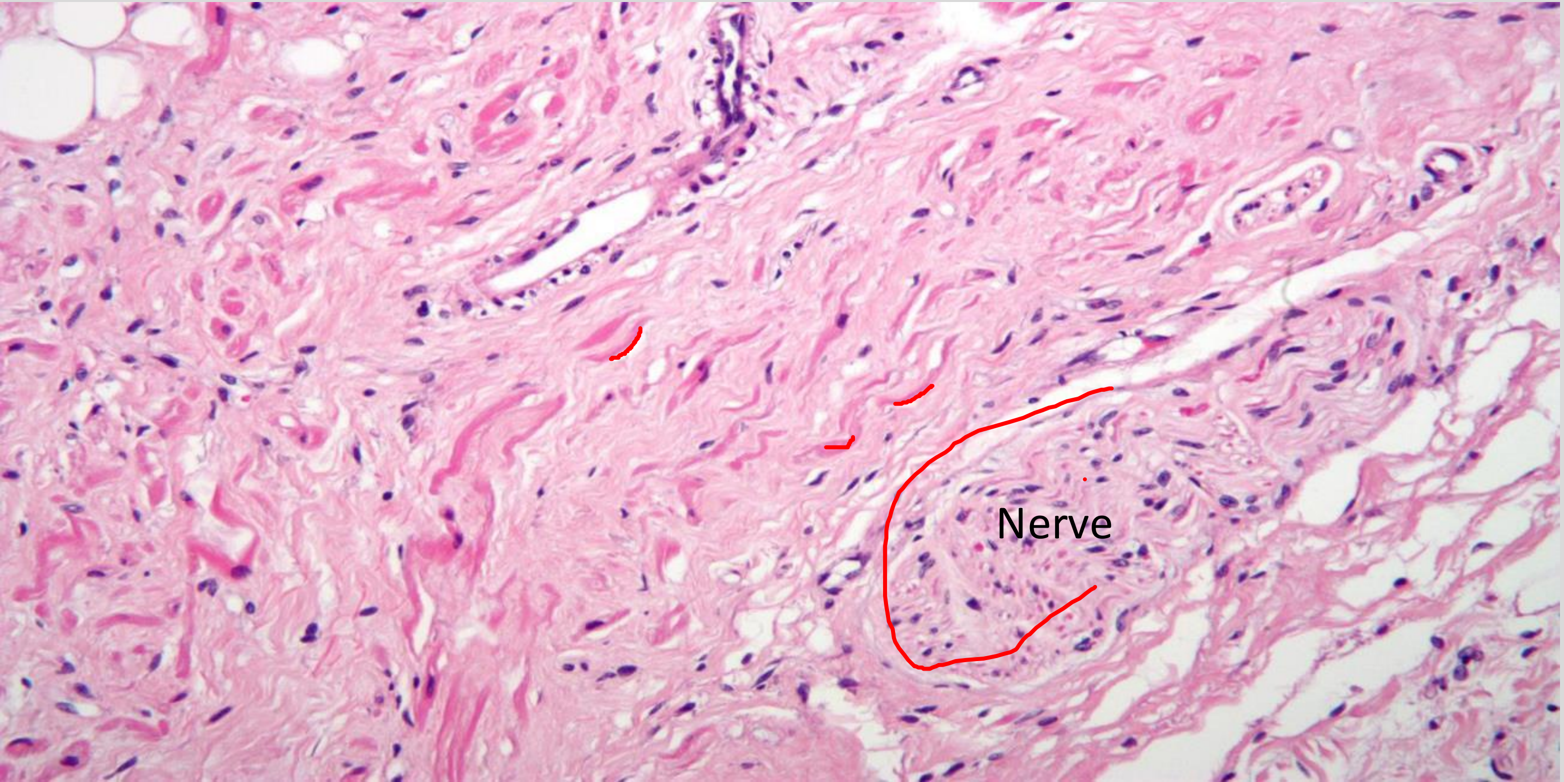
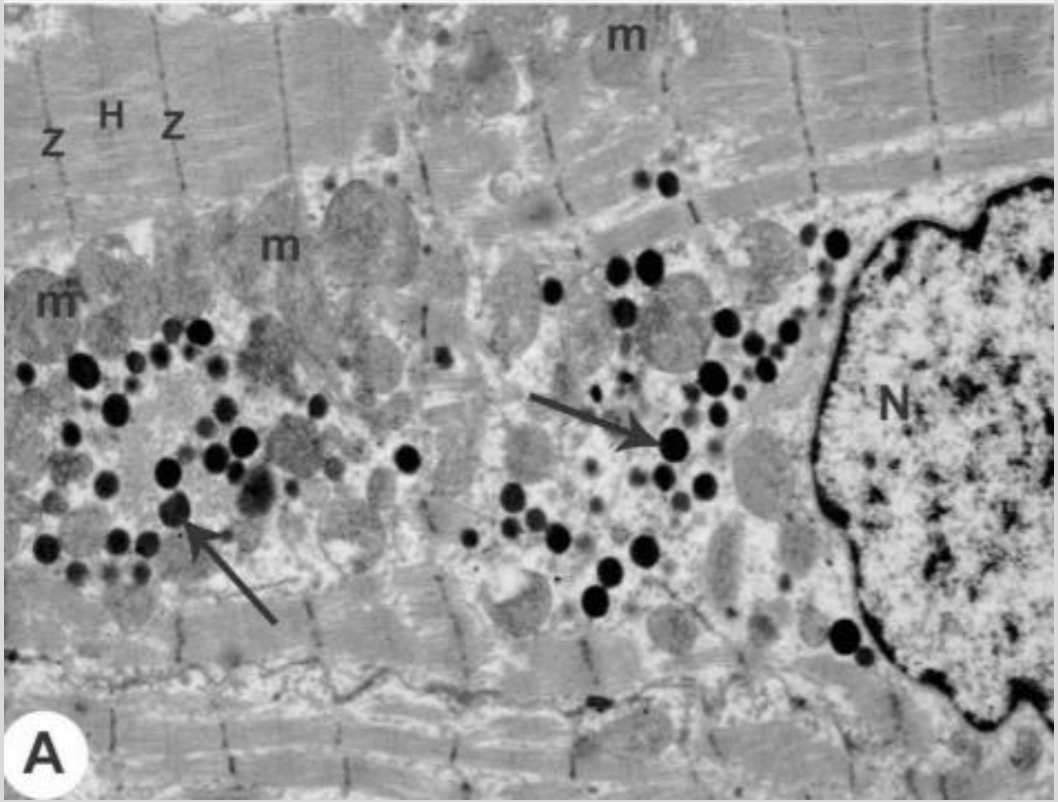
Nodal cells are wavy red fibers



Source: https://commons.wikimedia.org/wiki/File:Sinoatrial_node_high_mag.jpg#/media/File:Sinoatrial_node_high_mag.jpg

Atrial Myocyte



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ANP granules accumulate at nuclear poles
Note: BNP is produced primarily in ventricles, especially in heart disease, but not stored (no granules in ventricular myocytes)

Source: RA Eid, Int J Morphol 36:1031, 2018

<u>Basics of cardiac pathology</u>

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Cardiac hypertrophy: In areas with viable myocardium, pressure overload leads to wall thickening from ↑ myocyte diameter. Chamber dilatation is due to ↑ myocyte lengthening from series sarcomere addition. So, physiological and pathological changes in ventricular anatomy are generally reflected by changes in myocyte shape. *AM Gerdes, 2002; J Cardiac Failure*

Myocardial fibrosis: <u>*Interstitial fibrosis*</u> between myocytes contributes to ventricular stiffness and impaired relaxation. Collagen has the tensile strength of steel! *Myocardial infarction* results in <u>*myocyte necrosis with replacement fibrosis*</u>. This looks like a large area of fibrosis where myocytes should be. Hypertension leads to <u>*perivascular fibrosis*</u> around arteries and contributes to vessel stiffness.

Heart: Low power view

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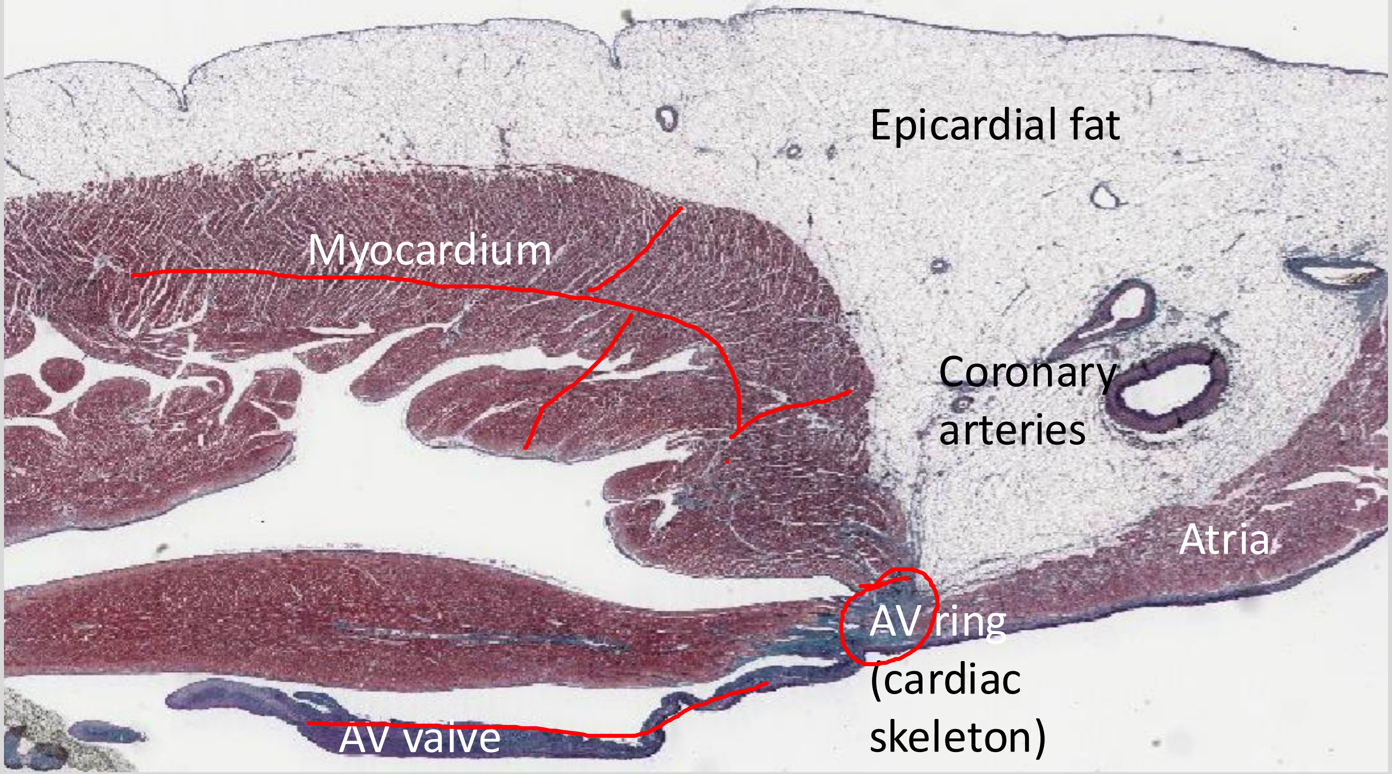

Source: Univ Michigan virtual histology

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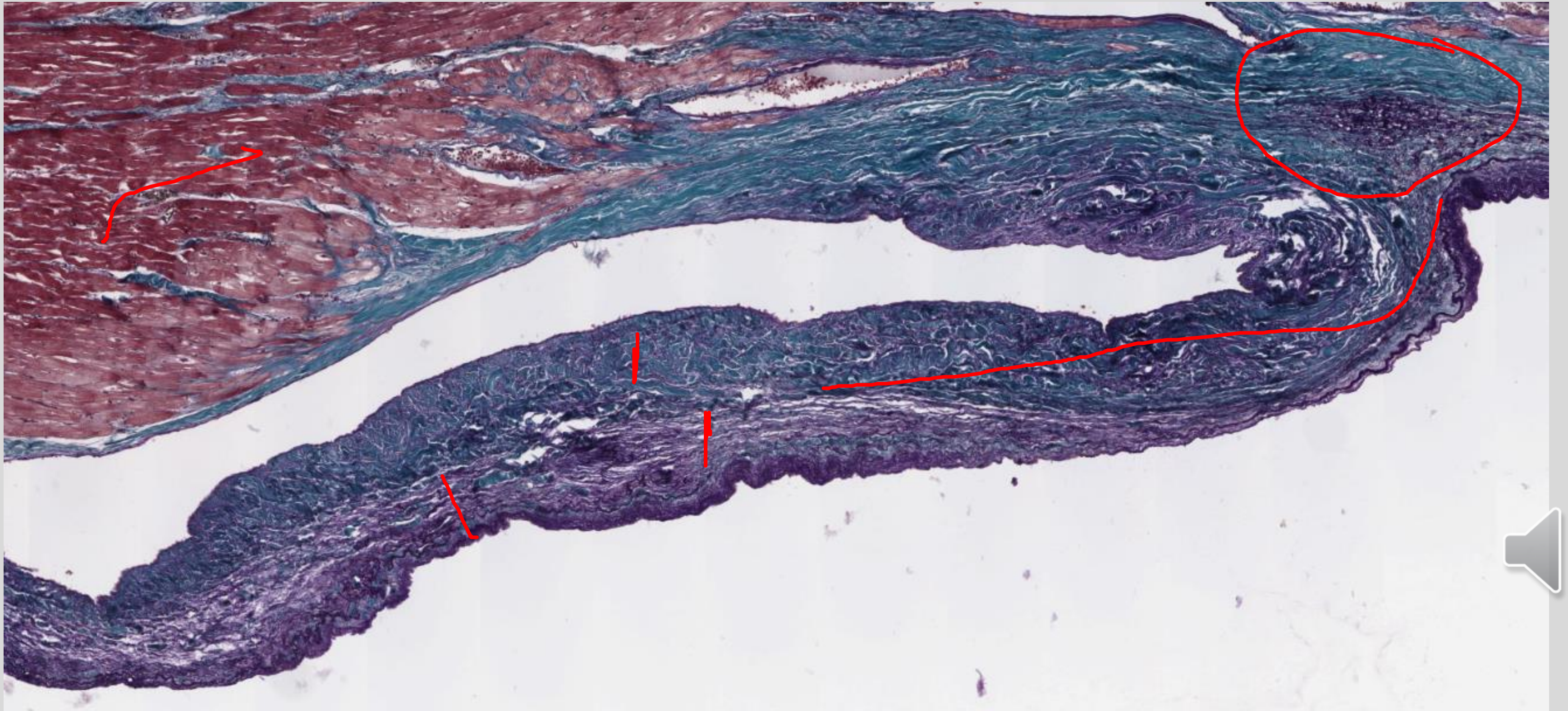

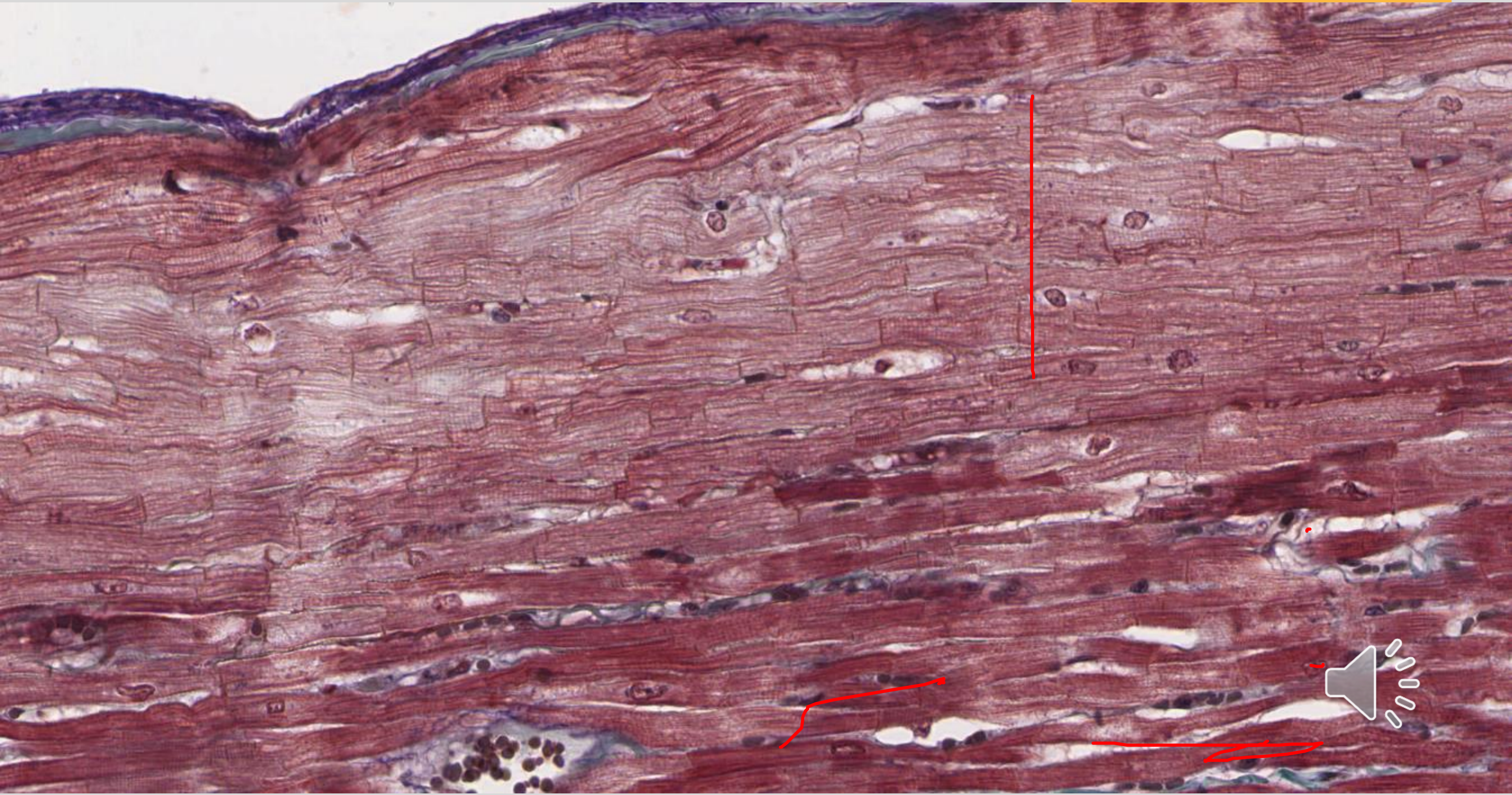

Histology and Virtual Microscopy Learning Resources University of Michigan

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https://histologyslides.med.umich.edu/Histology/Cardiovascular%20System/098-N_HISTO_40X.htm

Become familiar with heart histology by scanning Slide 98, virtual microscope: ventricle on left, atria on right, epicardium superior, AV valve inferior

Summary Slide

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- <u>Students should</u>: have a good mental image of the three layers of the heart and ventricular/atrial differences.
- Be able to describe how the morphologic composition of cardiac valves is quite functional.
- Be able to explain the basic morphology of the cardiac conduction system.
- Be able to identify various components of the cardiac skeleton and how the their histological composition underlies function.
- Be able to describe the basic structure of the pericardium and how this composition underlies its function.

Lecture Feedback Form:

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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R